

DAMON [1]-based {C,G,X}PU [un]attached NUMA Pages Migration

[1] Data Attributes Monitoring and Operations eNgin

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LINUX

PLUMBERS CONFERENCE

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Last Year, We Discussed

- DAMON supports memory tiering
- Next: general NUMA support
- Blocked by accessor obliviousness
- Idea/Plan for unblocking
(accessor-aware DAMON)

This Year, We Will Discuss

- Progress on accessor-aware DAMON
- Plan to implement DAMON for NUMA
 - User-space controlled version
 - Just-work version

DAMON in Nutshell

DAMON in One Sentence

“DAMON is a Linux kernel subsystem for efficient access monitoring and access-aware system operations.”

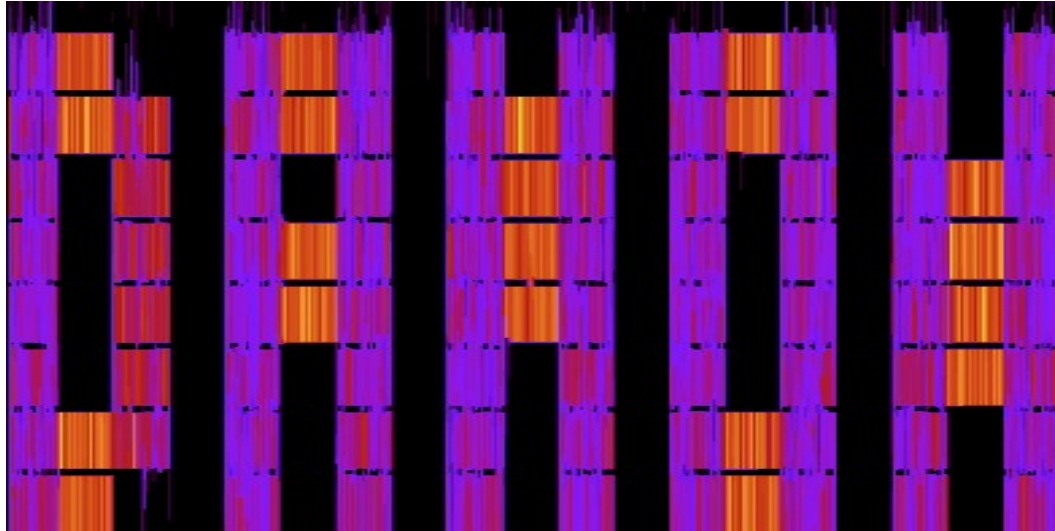


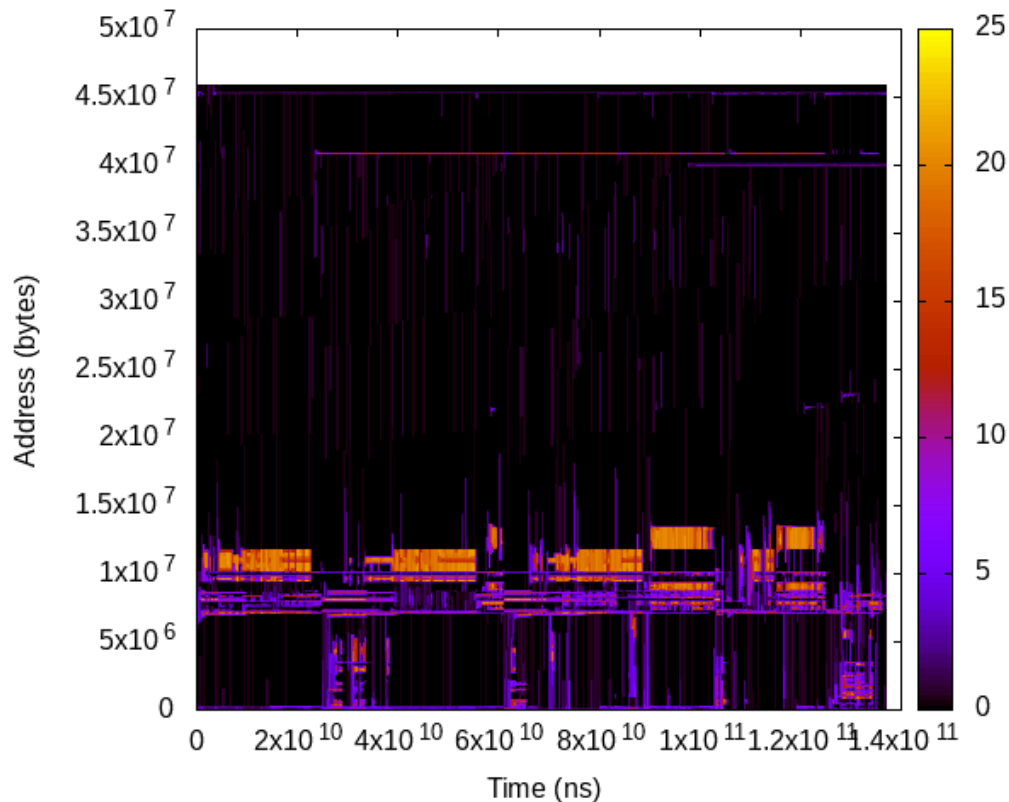
Image is generated by DAMON

High Level Implementation

- Spawn a kernel thread that does
- See if each page is accessed, every 5ms
- Inform users the findings, every 100ms
- Time intervals can be [auto-]tuned

Providing Access Information

- Location
- Frequency
- Stability
- Recency



Access pattern heatmap of
Splash2x/Raytrace

Design

- Goals: Accurate, Lightweight, Tunable and Automated
- Mechanisms: Sampling, Adaptive regions, Sysfs interface, Feedback loops
- Results
 - $<0.1\%$ CPU overhead on products

Availability and Usages

- Available on ≥ 5.15 upstream kernels
- Enabled on most [distro](#) kernels
- Being used for products and researches

DAMOS: Second Face of DAMON

- DAMon-based Operation Schemes
- Turn DAMON into access-aware memory management system operations engine
 - “Page out cold memory”
 - “Migrate hot/cold pages to DRAM/CXL node”

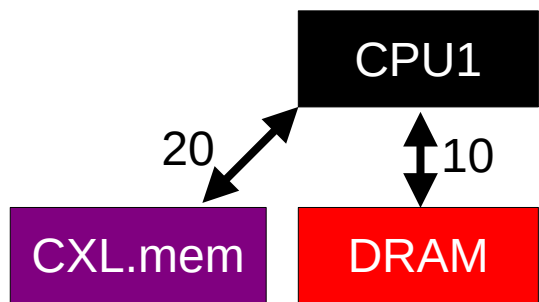
Community

- Public channels
 - Mailing list: damon@lists.linux.dev
 - Project website: <https://damonitor.github.io/>
- Private channels
 - Maintainer email: sj@kernel.org
 - DAMON Beer/Coffee/Tea [Meetup](#)

Background

DAMON Solves Memory Tiering

- Migrate hot/cold data to upper/lower tier
- Known users: SK hynix and Micron
- Community supports auto-tune [solution](#)



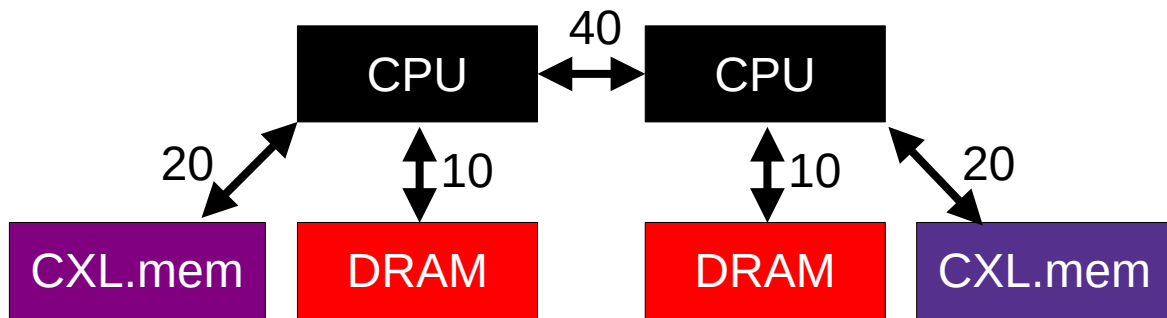
Possible tiered topology

	CPU
DRAM	10
CXL.mem	20

Relative access speed
from CPU to memory

Next: General NUMA Support

- “Migrate socket X-hot/cold pages to socket X-local/remote NUMA memory”
- Tiering for per-accessor migration path



Possible general NUMA topology

	CPU1	CPU2
DRAM1	10	50
CXL.mem1	20	60
DRAM2	50	10
CXL.mem2	60	20

Relative access speed
from CPU to memory

Problem: Accessor Unawareness

- DAMON lacks accessor info
 - Knows only “someone” accessed the data
- Can't find socket X-hot/cold data



B: Has... this data accessed?

K: What do you mean,
from CPU X or GPU Y?

B: Well... I don't know...

Accessor-aware DAMON

DAMON in One Slide

- Spawn a kernel thread that does
- See if each page is accessed, every 5ms
- Inform users the findings, every 100ms
- Time intervals can be [auto-]tuned

DAMON in One Slide

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Page Table Accessed Bit (A-bit)

- DAMON's *only* access check primitive
- S/W (DAMON) unset; H/W set for access
- No Accessor Info

Problem: A-bit Dependency

- DAMON is limited by its only single primitive
- Idea: add primitives
- Further: Why only access check primitive?

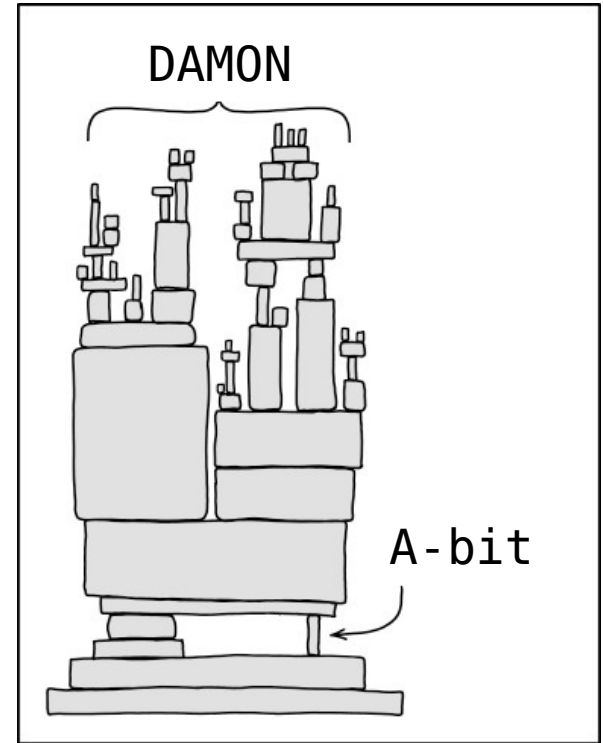


Image is retrieved from
<https://imgs.xkcd.com/comics/dependency.png>

Data Access → Data Attribute

- A-bit of the data is set
- CPU X executed load/store for the data
 - Cycle/Instruction sampling can be used
- CXL HMU reported the data as hot
- Data belongs to cgroup X

Data Attrs Monitoring

- Users specify attributes of interest
- Spawn a kernel thread that does
- See if each page has the attrs, every 5ms
- Inform users the findings, every 100ms
- Time intervals can be [auto-]tuned

Data Attrs Monitoring

- Users specify attributes of interest
- Spawn a kernel thread that does
- See if each page has the attrs, every 5ms
- Inform users the findings, every 100ms
- Time intervals can be [auto-]tuned

Perf Event: New Primitive

- Abstraction for generic perf monitoring
- Supports CPU cycle/instruction sampling
- Can be the accessor-aware primitive

Beyond A-bit Roadmap

- Extend DAMON for Data Attributes
 - Milestone 1, ETA: LPC'26
- Support perf event based access attribute
 - Milestone 2, ETA: LSFMMBPF'27
- Support AMD IBS, Intel PEBS, ARM SPE, ...

Milestone 1: Complete

- DAMON is (almost) renamed to DAMON
 - Data Attrs Monitoring and Operations eNgin
- DAMOS update is in progress

Milestone 2: In Progress

- Perf event based access monitoring
 - Akinobu suggested, SJ is revisioning
- Decode perf sample load/store instruction

Milestone 3: To Be Implemented

- Working prototypes exist
 - Ravi: AMD IBS, Intel PEBS Support
 - Kunwu: Arm SPE Support

Post-unblock Plan

User-controlled NUMA Support

- Requisites: Only accessor-aware DAMON
- Control w/ `sysfs` or user-space tool (`damo`)

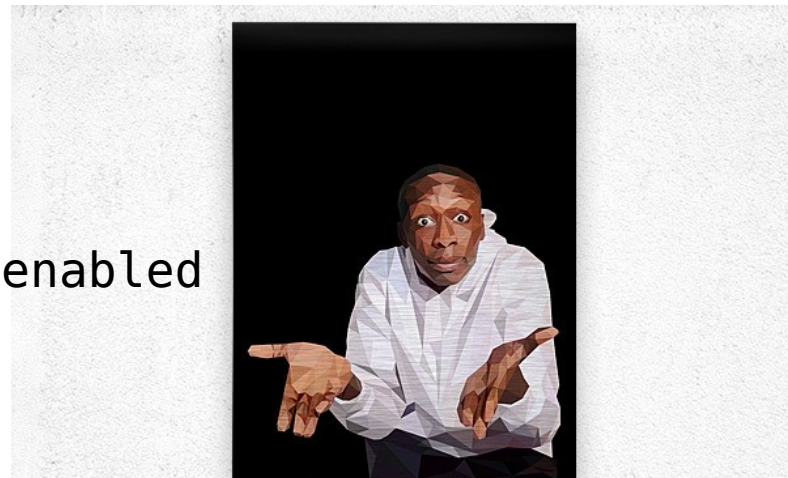


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<https://i.pinimg.com/474x/5e/01/44/5e0144f0d5d56db1ac11e19c674ab985.jpg>

NUMA Support That Just Works

- Dedicated built-in module: DAMON_NUMA
- Minimum tunables
- Allows flexible control (DAMON-X)

```
# echo Y > \  
/sys/module/daemon_numa/parameters/enabled
```



Wrapup

- Accessor-aware DAMON is cooking
 - ETA: LSFMMBPF' 2027
- User-control DAMON_NUMA will arrive
 - ETA: LPC' 2027
- DAMON_NUMA that just works is planned

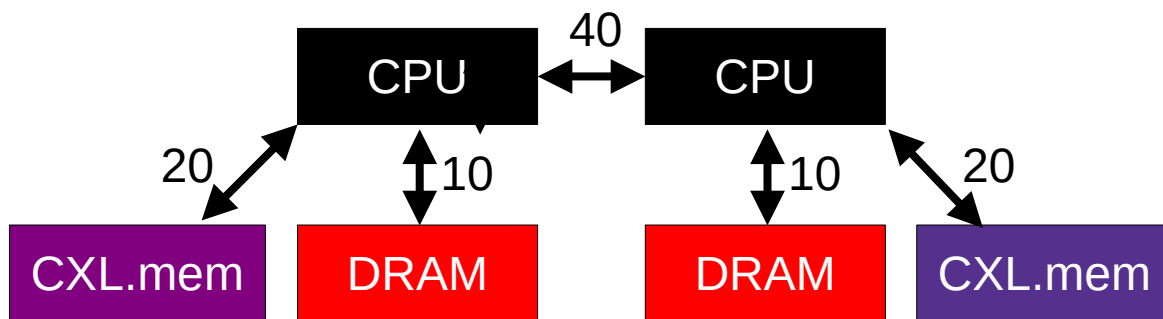
Discussion Time!

- Feel free to continue after this session on
 - Hallway, evening events
 - damon@lists.linux.dev
 - DAMON Beer/Coffee/Tea [chats](#)
 - sj@kernel.org

Backup Slides

NUMA for Dummies

- Access speed of memory depends on
 - Accessor (CPUs, GPUs, or ?PU),
 - Memory address, etc (e.g., compression rate)



Possible NUMA topology

	CPU1	CPU2
DRAM1	10	50
CXL.mem1	20	60
DRAM2	50	10
CXL.mem2	60	20

Relative access speed
from CPU to memory

DAMON for Memory Tiering

SK Hynix: Capacity Expansion

- Migrate hot pages CXL → DRAM
- Migrate cold pages DRAM → CXL
- Hot/cold threshold: tuned for their case
- A part of SK Hynix HMSDK
- ~94% llama.cpp [speedup](#)

Meta: TPP-DAMON

- DAMON-based **TPP** idea implementation
- Auto-tune hot/cold thresholds for TPP goal
 - Put as many hot data as possible in upper tier
- ~4.42% Taobench score **increase**

Micron: Dynamic Interleaving

- Change interleaving weights in runtime
 - For dynamic access pattern changes
- ~25% benchmark **speedup**

TPP-DAMON in More Details

- The idea is simple and scalable
 - Promote hot pages to the upper node, aiming high utilization of the upper node
 - Demote cold pages to lower node, aiming headroom free space of the node
- Applicable to N tiers

TPP-DAMON in More Details

```
damo start \  
  --numa_node 0 --monitoring_intervals_goal 4% 3 5ms 10s \  
    --damos_action migrate_cold 1 --damos_access_rate 0% 0% \  
    --damos_apply_interval 1s \  
    --damos_quota_interval 1s --damos_quota_space 200MB \  
    --damos_quota_goal node_mem_free_bp 0.5% 0 \  
    --damos_filter reject young \  
  --numa_node 1 --monitoring_intervals_goal 4% 3 5ms 10s \  
    --damos_action migrate_hot 0 --damos_access_rate 5% max \  
    --damos_apply_interval 1s \  
    --damos_quota_interval 1s --damos_quota_space 200MB \  
    --damos_quota_goal node_mem_used_bp 99.7% 0 \  
    --damos_filter allow young \  
    --damos_nr_quota_goals 1 1 --damos_nr_filters 1 1 \  
  --nr_targets 1 1 --nr_schemes 1 1 --nr_ctxs 1 1
```

TPP-DAMON in More Details

damo start \

Dear DAMON user-
space tool,

Start DAMON in
kernel.

```
--numa_node 0 --monitoring_intervals_goal 4% 3 5ms 10s \
--damos_action migrate_hot 0 --damos_access_rate 0% 0% \
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TPP-DAMON in More Details

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--nr_targets 1 1 --nr_schemes 1 1 --nr_ctxs 1 1
```

Create one DAMON worker thread, monitoring for node 0 (upper tier)

With auto-tuned monitoring intervals.

TPP-DAMON in More Details

```
damo start \
--numa_node 0 --monitoring_intervals_goal 4% 3 5ms 10s \
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--damos_filter allow young \
--damos_nr_quota_goals 1 1 --damos_nr_filters 1 1 \
--nr_targets 1 1 --nr_schemes 1 1 --nr_ctxs 1 1
```

And ask it to *migrate* un-accessed (access_rate 0%)
memory to node 1 (lower tier), *coldest* one first.
IOW, **demote** cold memory.

TPP-DAMON in More Details

```
damo start \  
  --numa_node 0 --monitoring_intervals_goal 4% 3 5ms 10s \  
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    --damos_nr_quota_goals 1 1 --damos_nr_filters 1 1 \  
  --nr_targets 1 1 --nr_schemes 1 1 --nr_ctxs 1 1
```

And do that once per second.

TPP-DAMON in More Details

```
damo start \  
  --numa_node 0 --monitoring_intervals_goal 4% 3 5ms 10s \  
    --damos_action migrate_cold 1 --damos_access_rate 0% 0% \  
    --damos_apply_interval 1s \  
    --damos_quota_interval 1s --damos_quota_space 200MB \  
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  --damos_nr_quota_goals 1 1 --damos_nr_filters 1 1 \  
  --nr_targets 1 1 --nr_schemes 1 1 --nr_ctxs 1 1
```

Within *one second*, do the migration only up to *200 MiB* of memory.

TPP-DAMON in More Details

```
damo start \  
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  --nr_targets 1 1 --nr_schemes 1 1 --nr_ctxs 1 1
```

Under the 200MiB/1s hard limitation, auto-tune the real soft limitation to make node 0's free memory ratio becomes 0.5% soon.

TPP-DAMON in More Details

```
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    --damos_quota_goal node_mem_free_bp 0.5% 0 \  
    --damos_filter reject young \  
  --numa_node 1 --monitoring_intervals_goal 4% 3 5ms 10s \  
    --damos_access_rate_hot 0 --damos_access_rate_cold 5% max \  
    --damos_filter allow young \  
    --damos_nr_quota_goals 1 1 --damos_nr_filters 1 1 \  
  --nr_targets 1 1 --nr_schemes 1 1 --nr_ctxs 1 1
```

Btw, if the page is PG_Young, don't do the migration.
DAMON might identify young page as not accessed.

TP

damo

Create another DAMON worker thread that monitors *node 1* (lower tier) with auto-tuned monitoring intervals.

Ask it to migrate memory shown $\geq 5\%$ access rate to *node 0* (upper tier), *hottest* first. IOW, **promote** hot pages.

Do the migration once per second, up to 200 MiB of memory.

Under the *200 MiB/s* hard limit, adjust the real soft limit aiming *node 0* memory utilization ratio 99.7%.

Btw, to that only if the page is *PG_Young*.

```
--damos_... node_mem_tree_bp 0.5% 0 \
--damos_filter reject young \
--numa_node 1 --monitoring_intervals_goal 4% 3 5ms 10s \
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--damos_apply_interval 1s \
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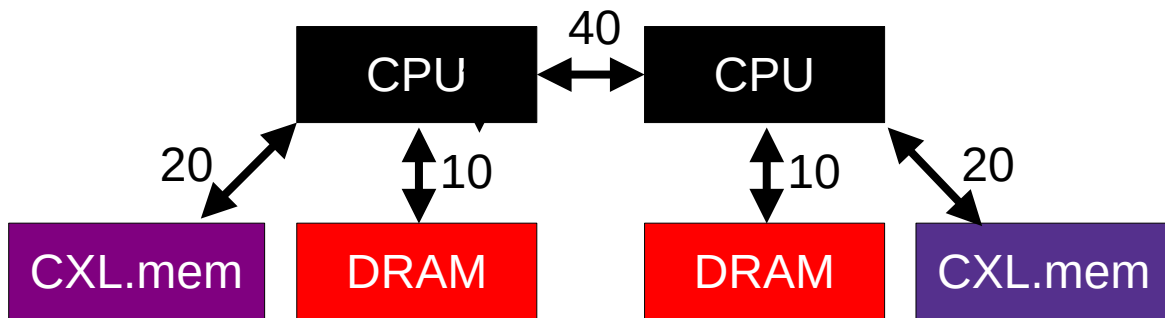
DAMON for Holistic NUMA Page Migration

Idea is Simple

- Tiering: special case NUMA management
 - All CPU [un]attached nodes
 - Single promotion/demotion path
- General NUMA: N Accessors
 - CPU, GPU, *PU
 - N promotion/demotion paths
 - Idea: run accessor-aware TPP for each path

Idea in Example

- TPP1 migrate CPU1-hot/cold memory
 - On DRAM1 → CXL.mem1 → DRAM2 → CXL.mem2 hierarchy
- TPP2 migrate CPU2-hot/cold memory
 - On DRAM2 → CXL.mem2 → DRAM1 → CXL.mem1 hierarchy



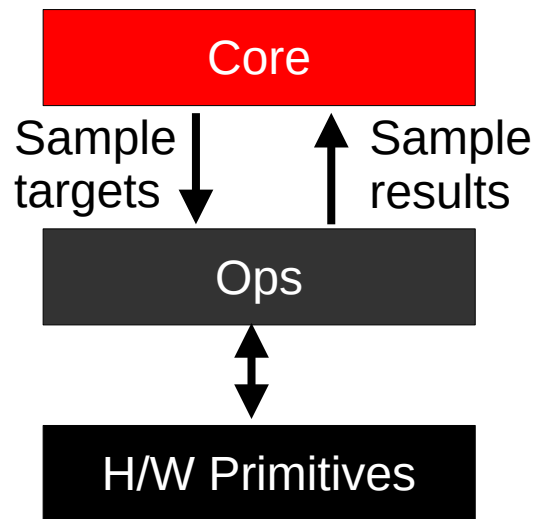
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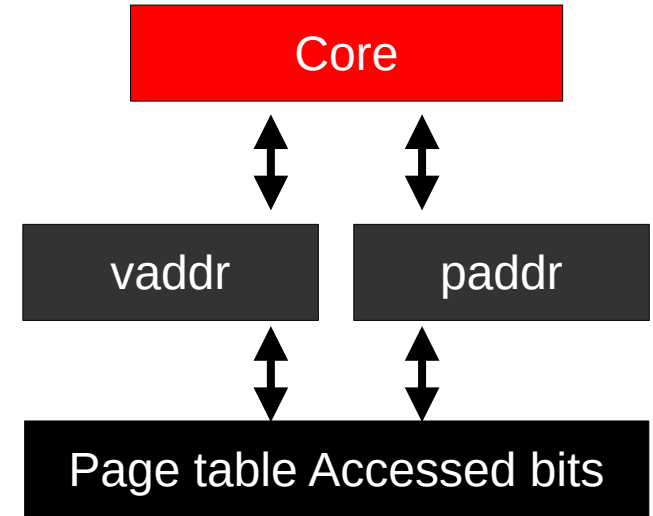
DAMON Core and Ops Layers

- Core: For pure software logic
 - Adaptively find pages to check access for
 - Assemble small samples to accurate pattern
- Ops: For h/w specific primitives
 - Check if core-specified samples got accessed
 - Using their favorite primitives
 - DAMON callers can implement their own ops



Existing DAMON Op Impls

- ‘vaddr’ and ‘paddr’
 - For virtual/physical address spaces monitoring
 - Both use page table Accessed bits
 - Lacks accessor CPU info



damon_report_attrs()

- Let any kernel components (e.g., perf event, GPU driver, etc) report their observed access information to DAMON core with single function call